THE PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

KILWA DISTRICT COUNCIL

FORM THREE TERMINAL EXAMINATION

PHYSICS

Time: 3 Hours **ne, 2026**

INSTRUCTIONS

- This paper consists of Section A, B and C with total of eleven (11) questions
- Answer all questions in section A and B and two questions from Section C
- Non-programmable calculator and mathematical table may be used
- Where necessary the following constants may be used
  - i. Acceleration due to gravity $=10 m/s^2$
  - ii. Specific heat capacity of copper = $390 J/kg^2$
  - iii. Specific heat capacity of piece of metal = 400J/kg °C
  - iv. Specific heat capacity of water = 4200J/kg °C
  - v. Density of mercury = $13600\ kg/m^3$
  - vi. Pie, $\pi = 3.14$.

SECTION A (16 MARKS)

1. Choose the most correct answer from the given questions and write its letter besides the number item in your answer booklet

   i. Suppose you are provided with a Coca-Cola, what is the name of the instrument you will use to measure diameter of the bottle's neck?

   A: Vernier caliper   C: Micrometer screw gauge
   B: Meter rule   D: Tape measure   E: Ruler

   ii. Which of the following does not affect surface tension?

   A: Impurities   B: Detergent
   C: Temperature   D: Volume of liquid   E: Cohesive force

   iii. The initial volume of a liquid in a burette was 57 cm^3. If X cm^3 of the liquid was poured out of the burette and the final volume of the liquid remaining was 23 cm^3, calculate the value of X.

   A: 80 cm^3 C: 34 cm^3
   B: 50 cm^3 D: 46 cm^3 E: 48 cm^3

   iv. Classify the force experienced when a metal solid or a hard object is twisted.

   A: Compression   C: Restoring
   B: Stretching   D: Torsion   E: Friction

   v. How much heat energy is given out by a copper block of 40 g mass when it cools from 140°C to 40°C?

   A: 20050 J   C: 2800 J
   B: 28080 J   D: 28000 J   E: 20800 J

vi. Which action will increase the density of a substance?

A: Increasing the mass of the substance while increasing its volume
B: Increasing the volume of the substance while the mass is kept constant
C: Decreasing the mass of the substance while decreasing its volume
D: Increasing the mass of the substance while fixing its volume
E: Increasing the mass of the substance while decreasing its volume

vii. If after a small displacement, a body retains its initial position, what type of equilibrium is this?

A: Stable equilibrium C: Neutral equilibrium
B: Unstable equilibrium D: Natural equilibrium E: Bistable equilibrium

viii. Suppose an engine raises 200 kg of water steadily through a height of 60 m in 20 second. The upward force used is equal to the weight of water raised. Calculate the power in kW.

A: 6 Kw C: 5 kW B: 3 kW D: 7 kW E: 4 kW

ix. The following statements about magnetic lines of force are correct except

A: Always form close loops
B: Start at north pole and end at the south pole
C: Cross one another
D: Strong where the lines are closer together
E: Pass through all materials both magnetic and non-magnetic

x) Why windmills are constructed around the coastal areas, in open planes, in gaps of mountain range or at the top of rounded hills?

A: Because these are cheap areas to harvest wind energy
B: Because there are no obstruction and strong winds blow to rotate wind turbine propellers
C: Because when the sun heats the atmosphere some patches become warmer than the others
D: Because there is less movement of people and vehicle in those areas
E: Because the wind turbines are noise and can spoil the landscape

QN	I	II	III	IV	V	VI	VII	VIII	IX	X
ANSWER										

2. Match the descriptions provided in List A with their corresponding answers in List A to make meaningful statements

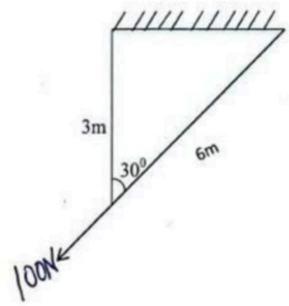
LIST A	LIST B
i. Is useful in drawing water from a borehole	A. Wheel barrow
ii. Applies the principle of class lever	B. Wheel and axle
iii. Its velocity ratio increases by increase the length of the ramp	C. Hydraulic press
	D. Tong
iv. The effort is between the load and the fulcrum	E. Claw hammer
v. Operates based on the principle of pressure in liquids	F. Inclined plane
vi. Carries objects from one point to another horizontally	G. Pulley
	H. Windmill
	I. Screw jack

QUESTIONS	I	II	III	IV	V	VI
ANSWERS						

SECTION B (54 MARKS)

3. a) Differentiate vector quantity from scalar quantity

b) Two ropes of 3M and 6M long pulled by a force of 100N as shown below. Find the tension in each rope if they make angle 30° between them

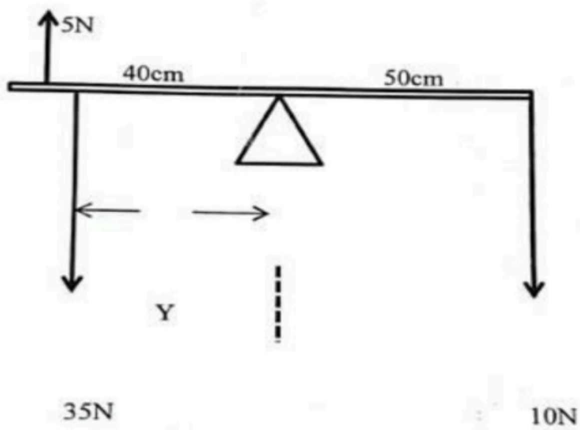


4a) A hot air balloon including the envelope gondola burner and fuel and one passenger has a total mass of 450g, The air outsides the ballon is 20°C and has density of 1.29kg/m³. The Air inside the envelope is heated to a temperature of 1200°C at which it has density of 0.9kg/m³. What volume must envelope expand in order to lift the balloon into the Air

b) An object floats in water with 60% of its of its volume submerged, if the object was placed in method with density of 0.79g/cm³. What percentage of its volume would be submerged

5a) A scale pan weight of 0.4N was attached on a spring balance and produces an extension of 24mm when the load of 2N was placed on it. Calculate the load on the scale pan when the extension is 16mm

b) The diagram below represents a meter rule pivoted at its centre, using the value of the forces and distances shown in the diagram. Calculate the distance Y if the system is in equilibrium



6. A) Why particles in solid expand when left in hot water? Briefly explain using kinetic theory of matter

b) A piece of metal with mass 200g at temperature of 100°C is quickly transferred into 50g water at 20°C. Find the final temperature

7a) why a big elephant manages to walk comfortably in muddy soil without sinking while a Human being may sink easily

b) Draw and explain well labelled diagram that pressure depends on depth

8a) You are provided with two types of mirror, concave mirror and convex mirror. What type of mirror among the two will you prefer in driving car to the see the traffic at your back? Explain two reasons for your choise

b) An iron load is 600 M long at 0°C, it is mounted alongside a copper load and both are always maintained at the same temperature. When they are heated to 100°C, it is found that the difference in their length is the same as it was 0°C. What is the length of the copper load at 0°C?

SECTION C (30 MARKS)

Answer only two questions

9. a) Mention two demerits of Anomalous expansion

b) A pieces of copper is dropped into water if the temperature of the water is rising. What is happening to the copper?

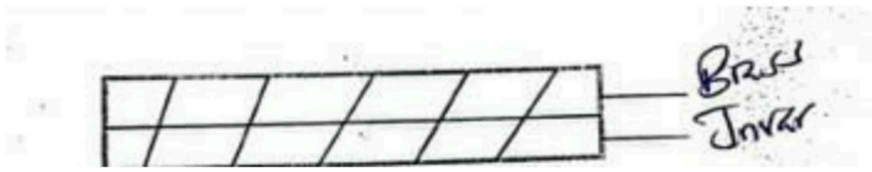
c) A fixed mass of a gas has a volume of 1.25 litres at pressure of 76.0cm of mercury. Find temperature of 27.0°C. The gas expands to volume of 1.55 litres raising the pressure to 80.00cm of mercury. What is the final temperature of the gas in °C

10 a) Differentiate Heat from temperature

b) i) Why electrical cable are left sagging during installation

ii) An inflated balloon hung in the open at wedding burst when the temperature of the environment rises

c) A fixed below shows a brass bimetallic strip at room temperature

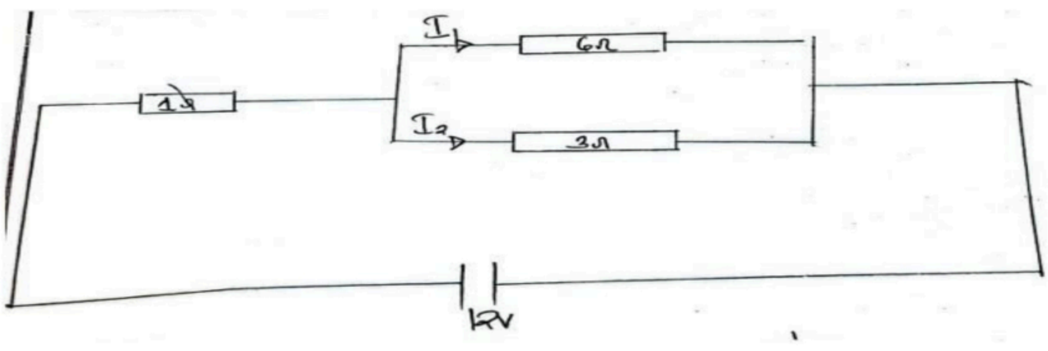


Given that brass expands more than invar when both are heated equally. Sketch the appearance of the strip after being cooled to several degree below room

11. a) Three capacitors with capacitance of 15 μF, 25μF and x are configured in a circuit such that the equivalent capacitance is 7μF

i. Identify the type arrangement of capacitors
ii. Determine the capacitance of capacitors

b) Study the electrical arrangement in the figure below and then answer the questions that follows



i. Calculate the current flowing in each of the three resistors
ii. Find the potential difference across each resistor

PHYSICS FORM THREE MARKING SCHEME

1. @1 = 1

I	II	III	IV	V	VI	VII	VIII	IX	X
A	C	C	D	E	B	C	A	C	B

2. @1 = 1

I	II	III	IV	V	VI
G	E	F	A	C	H

3.

a/

~~VECTOR~~ SCALAR QUANTITY	~~SCALAR~~ VECTOR QUANTITY
i/ Are those physical quantity which have only Magnitude	i/ Are the physical quantities which have both Magnitude and direction
ii/ They change if magnitude change	ii/ They change either if Magnitude or direction or both changes
iii/ They are specified by a magnitude and unit only	iii/ They are also specified by a number along with the direction and unit
iv/ They obey the ordinary Laws of Algebra	iv/ They don't obey the ordinary Laws of Algebra.

4 Marks.

b/ Solution

Length for rope one $L_1 = 3m$

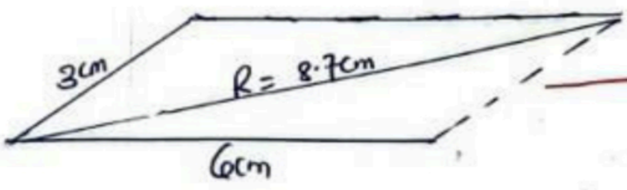
Length for rope Two $L_2 = 6m$ —— 01

$\theta = 30°$

Weight/force $= 100N$

By Parallelogram Law.

1cm represent 1m.



—— 01

1

From.

$100N = 8.7m$.

$x = 3m$

$8.7mx = 100N \times 3m$ —— 1/2

$\frac{8.7mx}{8.7m} = \frac{300Nm}{8.7m}$.

$x = 34.48N$

Tension in rope 3m is 34.48N —— 1/2

Length to $L_2 = 6m$

$100N = 8.7m$. —— 1/2

$x = 6m$.

$8.7mx = 600Nm$

Tension in rope 6m is 68.9N —— 1/2

Qn14

a/ Data.

Mass of ballon and Load $= 450kg$

Density of air $1.29kg/m^3$ at $20°C$ —— 01

Density of agas $0.90kg/m^3$ at $120°C$ $\rho = m/v$

from.

$$Volume = \frac{\text{Mass of Load}}{\text{Density of air} - \text{Density of agas}}$$ —— 01

$$V = \frac{450kg}{1.29kg/m^3 - 0.90kg/m^3}$$

$V = 1153.846cm^3$. —— 03

b/ Data.

Fraction Volume of submerged in water $= 60\%$

Density of methanol $= 0.79g/cm^3$.

Percentage volume of submerged of methanol = ? —— 1/2

$$\text{Fractional volume submerged} = \frac{\text{Density of object}}{\text{Density of water}}.$$

$$60\% = \frac{\rho_s}{\rho_w}.$$

$$60\% = \frac{\rho_s}{1g/cm^3}.$$ —— 1/2

$\rho_s = 60\% \times 1g/cm^3$

$\rho_s = 0.6g/cm^3$. —— 01

4b/ Fractional Volume submerged $= \frac{\text{Density of object}}{\text{Density of fluid}} \times 100\%$ —— 1/2

$= \frac{0.6\,g/cm^3}{0.79\,g/cm^3} \times 100\%$ —— 1/2

$= 0.75 \times 100\%$

$= 75\%$

The percentage of the volume of methanol submerged is 75%. —— 01

Qn5

a/ Weight of scale pan $W_1 = 0.4N$

Extension $e_1 = 2.4mm \longrightarrow 0.024m$ —— 01

Weight of Load $W_2 = 2N$

From Hookes Law.

$F \propto e$.

$F = Ke$. —— 01

$k = F/e$.

$k = \frac{W_1 + W_2}{e} = \frac{2N + 0.4N}{0.024m}$.

$k = 100N/m$. —— 01

Load is on scale pan when the extension

$16mm = 0.016m$.

Let Load $= W_3$.

$k = F/e$. —— 01

$k = F/e$.

$k = \frac{W_1 + W_3}{e}$

$100N/m = \frac{0.4 + W_3}{0.016}$. —— 01

$100 \times 0.016 = 0.4N + W_3$

$1.6N = 0.4N + W_3$

$W_3 = 1.6N - 0.4N$

$W_3 = 1.2N$. —— 01

The Load on the scale pan when the extension is 16mm is 1.2N

5b/ Principle of Moments.

Sum of clockwise Moment = Sum of Anticlockwise ——— 01

$5 \times 40 + 10 \times 50 = 35 \times Y$

$700 = 35Y$

$Y = 700/35$ ——— 01

$Y = 20cm$

The distance of $Y = 20cm$. ——— 01

Qn6

a/ According to kinetic theory of matter all matter are made up of small particles which are in Random motion, When the temperature increase particle (molecule) gain heat energy which make the Molecules to Vibrate and the vibration of Molecules Making the solid object to expand ——— 03

b/ Data.

Mass of metal $200g \Rightarrow 0.2kg$.

Temperature of metal $= 100°C$

Mass of water $50g \Rightarrow 0.05kg$

θ_i of water $= 20°C$ ——— 01

$C_{metal} = 400J/kg°C$

$C_{water} = 4200J/kg°C$

Asked: Final temperature.

From

Heat Loss = Heat gain

$MC_m\Delta T = MC_w\Delta T$. ——— 01

$0.2 \times 400(100 - \theta f) = 0.05 \times 4200(\theta f - 20)$

$\frac{80}{\cancel{8}0}(100 - \theta f) = \frac{210}{\cancel{8}0}(\theta f - 20)$. ——— 01

$8 \times 100 - \theta f = 21/\cancel{8}(\theta f - 20) \times \cancel{8}$

$8(100 - \theta f) = 21(\theta f - 20)$.

$800 - 8\theta f = 21\theta f - 420$. ——— 01

$800 + 420 = 21\theta f + 8\theta f$

$\frac{1220}{29} = \frac{29\theta f}{29}$.

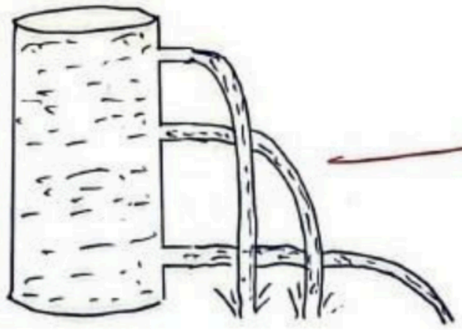
$\theta f = 42.06°C$.

The final temperature $= 42.06°C$. ——— 02

Q7

a/ The feet of an Elephant are large to distribute its significant weight over a large area thereby reducing the pressure exterted on the ground and prevent sinking — 06

b/



— 03

8.

@ The following are similarities between the human eye and the lens camera.

(i) Both Uses convex lens. (1½ mark @)

(ii) Both forms real, reduced and inverted Image. Total 4.5 marks.

(iii) The amount of light entering the camera is Controlled by the diaphragm while amount of light entering the eye is controlled by the Pupil.

(b) Solution

Data given

ho (height of object) = 1.8m

u (object distance) = 2.5m

focal length of lens = 0.05m

V (image distance) = ? 00½ marks

Magnification (m) = ?

height of image (hi) = ?

for a lens camera;

$$m = \frac{f}{u-f}$$

$$m = \frac{0.05m}{2.5m - 0.05m} = \frac{0.05m}{2.45m}$$

$\underline{m = 0.02}$ (Magnification) (02 marks)

Since $m = \frac{v}{u}$

$V = mu = 0.02 \times 2.5m$

$\underline{V = 0.05m}$ (Image distance) (01 marks)

but $m = \frac{h_i}{h_o} \Rightarrow h_i = mh_o$

$h_i = 0.02 \times 1.8m$

$\therefore$ $\underline{h_i = 0.036m \text{ or } 3.6cm}$. (01 marks)

(height of Image)

SECTION C

Qn 9

a/ i/ A glass bottle filled with water and sealed cracks if cooled in a deep freezer

ii/ If water freezes in a pipe, it may cause the pipe to burst open due to expansion ——— 03 Marks.

iii/ Weathering of rocks

b/ If the hot copper pan is dropped into a tube of water, temperature of copper pan is transfered to the water, this Means that the temperature of the copper pan is dropped at the temperature of water rises. ——— 06 Marks.

c/ Data.

$V_1 = 1.25L$

$V_2 = 1.55L$ ——— 01

$P_1 = 76.0 cmHg$

$P_2 = 80.0 cmHg$

$T_1 = 27°C + 273 = 300k$

$T_2 = ?$

$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$ ——— 01

$\frac{P_1V_1T_2}{P_1V_1} = \frac{P_2V_2T_1}{P_1V_1}$ ——— 01

$T_2 = \frac{P_2V_2T_1}{P_1V_1}$ ——— 01

$T_2 = \frac{80.0 \times 1.55 \times 300}{76.0 \times 1.25} = 391.6k$ ——— 01

$T_2 = 391.6 - 273 = 118.6°C$

$T_2 = 118.6°C$ ——— 01

Qn 10

a/

HEAT	TEMPERATURE.
i/ Is a form of energy possessed by a body due to its tempe rature change	i/ Is the degree of hotness or coldness of a body
ii/ SI unit is Joule	ii/ SI unit is Kelvin (k)
iii/ It cannot be Measured directly	iii/ It can be Measured by a thermometer
iv/ It can be transfered from one ~~form~~ body to another	iv/ It cannot be transfered ——— 03 Marks

b/ i/ Solid wires like Telephone electrical wires experience expansion and contraction due to changes in Temperature during summer and winter respectively to avoid breakage or snapping of wires due to Contraction In Winter the wires are let Loose between the poles ——— 03

ii/ If the pressure Inside the balloon gets too high, it will burst; the balloon is exposed to heat or direct sunlight, when the temperature rises, the Air or Helium within the balloon expands. ——— 03

c/ Bimetallic strip when heated it bend towards the iron this is becouse brass expand faster than ~~iron~~ Invar and become larger ——— 03

Bimetallic strip when cooled, Brass Contracts faster than Invar, the bimetallic strip therefore bends towards brass side. ——— 03

Qn 11

a/ $C_1 = 15\mu F$, $C_2 = 25\mu F$ $C_3 = x$.

Equivalent capacitor $= 7\mu F$.

Arrangement of capacitor is series becouse of equivalent Capacitance is less with the value of capacitor ——— 03

11(a)(ii) $\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$.

$\frac{1}{7} = \frac{1}{15} + \frac{1}{25} + \frac{1}{x}$ $\quad \frac{25x + 15x + 375}{375x}$

$\frac{95x}{95} = \frac{2625}{95}$ —— 03

$x = 27.63\mu F$

b) Parallel Arrangement

$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$

$\frac{1}{R_T} = \frac{1}{6} + \frac{1}{3}$ —— 01

$\frac{1}{R_T} = \frac{3+6}{18}$

$R_T = 2\Omega$. —— 01

In series

$R_T = R_1 + R_2$

$1\Omega + 2\Omega$

$R_T = 3\Omega$. —— 01

From ohms Law

$V = IR$. make I the subject

$I = V/R$

$I = 12/3 = 4A$ —— 01

Potential difference

$V_T = V_1 + V_2$.

$V_1 = I_1 R_1$ —— 01

$V_T = I_1 R_1 + V_2$.

$(4 \times 1) + V_2$.

$\cancel{V_2} = 12V = 4V + V_2$

$V_2 = 8V$. —— 01

Current in $I_1 = ?$

$I_1 = \frac{V_1}{R_1} = \frac{8V}{6}$

$I_1 = 1.33$

$I_2 = \frac{V_2}{R_2}$

$= 8/3$

$I_2 = 2.66.$

$R_1 = 1\Omega = 4A.$

$R_2 = 6\Omega = 1.33$

Potential difference in Each Resistor. $R\ 1\Omega$.

$V = 4A \times 1 = 4V.$

$V_1 = 4V.$

P.d in Parallel Resistors 6Ω and 3Ω are same which is 8V